

HUMAN-TRUST-AWARE BLOCKCHAIN SYSTEM FOR PREVENTING SILENT DATA MANIPULATION IN DIGITAL GOVERNANCE PLATFORMS

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ABSTRACT

Digital governance platforms manage sensitive public data and require strong guarantees of integrity, transparency, and accountability. However, centralized architectures remain vulnerable to silent data manipulation, particularly from insider threats where unauthorized modifications may go undetected. Although blockchain technology provides immutability and traceability, most existing solutions focus primarily on cryptographic protection and neglect the dynamic role of human trust in governance operations. This paper proposes a Human-Trust-Aware Blockchain System that integrates dynamic trust evaluation with a permissioned blockchain framework to prevent unauthorized data alterations. User trust scores are continuously updated based on behavioral patterns, access frequency, and role compliance, while smart contracts enforce trust-aware access control before approving sensitive operations. Suspicious activities trigger automated alerts and access restrictions. Experimental evaluation in a simulated governance environment demonstrates improved detection of insider misuse and enhanced data integrity compared to traditional centralized systems. The proposed framework strengthens transparency, accountability, and public confidence in digital governance services.

Keywords: Blockchain, Digital Governance, Human Trust Modeling, Data Integrity, Insider Threats, Cybersecurity.

I. INTRODUCTION

Digital governance platforms have become a foundational component of modern public administration, facilitating the efficient delivery of services such as digital identity management, land record systems, healthcare databases, taxation services, and online voting mechanisms. As governments increasingly digitize administrative processes, these platforms handle large volumes of sensitive citizen data, making data integrity, transparency, and accountability critical requirements. Any compromise in data authenticity can directly affect public service delivery, legal procedures, and public trust in governance institutions.

Most existing digital governance infrastructures are built upon centralized database architectures combined with hierarchical administrative control mechanisms. Although such systems simplify coordination and operational management, they introduce significant security vulnerabilities. Among these, silent data manipulation represents a particularly critical threat. Silent manipulation refers to unauthorized modifications of records that occur without immediate detection. Unlike conventional external cyberattacks, such manipulations frequently originate from insider entities possessing legitimate access privileges. Because these actions may not trigger alerts in traditional monitoring and logging systems, they can remain undetected for extended periods, leading to reduced transparency, erosion of public confidence, and potential legal or societal consequences.

Blockchain technology has emerged as a promising approach for addressing data integrity challenges due to its decentralized architecture, cryptographic hashing mechanisms, and immutable ledger structure. By ensuring that recorded transactions cannot be altered without generating verifiable evidence, blockchain enhances transparency and auditability. However, many existing blockchain-based governance solutions focus primarily on technical immutability and cryptographic validation. These approaches often assume honest behavior

among authorized participants and therefore overlook the dynamic and human-centric dimensions of trust within governance environments.

In practical governance systems, trust extends beyond cryptographic guarantees. It represents a socio-technical construct influenced by user behavior, administrative accountability, policy compliance, and operational consistency. Insider threats frequently arise not solely from technical vulnerabilities but from the misuse of legitimate authority. Consequently, securing digital governance platforms requires not only immutable data storage mechanisms but also adaptive and continuous evaluation of human behavior.



Figure 1: Digital Governance Security Challenges Illustrated

To address this limitation, this research proposes a Human-Trust-Aware Blockchain System that integrates dynamic trust modeling, behavioral analytics, and blockchain immutability within a unified framework. The proposed system continuously evaluates user trust levels based on operational patterns and enforces trust-aware access control through smart contracts. By combining cryptographic security with human-centric trust assessment, the framework aims to detect and prevent silent data manipulation while strengthening transparency, accountability, and public confidence in digital governance platforms.

II. LITERATURE REVIEW

Digital governance systems need strong security to keep data safe, accurate, and transparent. Traditional systems usually use centralized databases with access control and audit logs. However, these systems are still vulnerable to insider threats, where authorized users can secretly change important data. Such hidden changes reduce public trust and system reliability.

Blockchain technology has been proposed as a solution because it stores data in a decentralized and tamper-resistant way. Its cryptographic structure makes it very difficult to modify records once they are added. Many researchers have suggested using blockchain for land records, digital identity, and public services. While blockchain protects against external attacks, it often assumes that authorized users behave honestly and does not fully solve insider threats.

To detect insider threats, researchers have developed methods like monitoring user activity, analyzing behavior patterns, and using machine learning. These methods help identify suspicious actions. However, most of them depend on centralized monitoring systems, which can also be compromised.

Trust modeling is another approach where users are given trust scores based on their past behavior. If a user behaves suspiciously, their trust level decreases. Trust-based systems are used in cloud computing and IoT, but they are rarely combined with blockchain systems.

III. PROBLEM STATEMENT

Despite the adoption of advanced security technologies, digital governance platforms continue to face the risk of silent data manipulation due to excessive administrative privileges, limited continuous monitoring, and the absence of trust-aware decision-making mechanisms. There is a need for a system that not only ensures immutable data storage but also dynamically evaluates human behavior to detect and prevent malicious activities in governance environments.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed Human-Trust-Aware Blockchain System is designed to prevent silent data manipulation in digital governance platforms. It combines trust evaluation with blockchain technology to provide secure, transparent, and accountable data management. The system follows a layered architecture to ensure proper monitoring, validation, and protection of governance data.

The architecture consists of four main layers: the User Interaction Layer, Trust Evaluation Layer, Blockchain Integrity Layer, and Monitoring and Alert Layer.

The User Interaction Layer allows authorized users such as administrators and government officials to perform operations like creating, updating, verifying, and retrieving data. Every action is authenticated and recorded so that no operation bypasses the security system.

The Trust Evaluation Layer continuously monitors user behavior and assigns a dynamic trust score based on factors such as past activities, access patterns, and role compliance. Before any sensitive action is approved, the system checks the user's trust score. If the score is below the required threshold, access may be restricted or additional monitoring may be applied.

The Blockchain Integrity Layer ensures that all approved operations are securely recorded on a permissioned blockchain. Each transaction is time-stamped, cryptographically secured, and permanently stored. Smart contracts verify trust conditions before allowing data modifications. Once recorded, transactions cannot be altered, ensuring long-term data integrity.

The Monitoring and Alert Layer detects unusual activities by analyzing trust score changes and transaction behavior. If suspicious activity is identified, the system generates alerts and can automatically restrict access or escalate the issue to higher authorities.

Overall, the system workflow begins when a user initiates an operation. The trust score is verified, and if approved, the transaction is executed and recorded on the blockchain. This combined approach ensures both behavioral monitoring and immutable data protection, effectively reducing the risk of silent data manipulation.

V. HUMAN TRUST MODELING METHODOLOGY

Human behavior significantly affects the security of digital governance systems. Even with strong technical safeguards, insider misuse can lead to silent data manipulation if user activities are not continuously evaluated. To address this issue, the proposed system implements a dynamic human trust model that assesses the reliability of each authorized user in real time.

Each user is assigned a trust score based on behavioral and operational factors such as role compliance, access frequency, consistency of past actions, sensitivity of accessed data, and deviation from normal activity patterns. These parameters together provide a comprehensive measure of user trustworthiness.

An initial trust score is assigned during authorization and is continuously updated using a weighted aggregation method:

$$T_u = \sum_{i=1}^n (w_i \times P_i)$$

where T_u represents the trust score of user u , P_i denotes trust parameters, and w_i represents their respective weights, with the total weight equal to one.

Trust scores increase with consistent and compliant behavior and decrease when suspicious actions are detected. Based on predefined thresholds, users may receive full access, restricted access with monitoring, or access denial. These trust-aware decisions are enforced automatically through smart contracts within the blockchain layer.

By integrating dynamic trust evaluation with blockchain enforcement, the system ensures adaptive access control and strengthens protection against insider threats.

VI. BLOCKCHAIN-BASED DATA INTEGRITY MECHANISM

Digital governance platforms handle highly sensitive public records, making data integrity essential. Unauthorized modifications can lead to administrative errors, legal risks, and loss of public trust. To address this issue, the proposed system integrates a blockchain-based mechanism that ensures secure, transparent, and tamper-resistant data management.

Unlike centralized databases, blockchain maintains a distributed and immutable ledger. Once recorded, transactions cannot be altered or deleted without leaving verifiable evidence. This immutability makes blockchain well-suited for preventing silent data manipulation.

The proposed system employs a permissioned blockchain in which only authorized government entities participate. Every governance operation—such as data creation, modification, or approval—is recorded as a time-stamped and cryptographically secured transaction. Each record includes the encrypted user identity, operation type, timestamp, current trust score, and the hash of the related data. Any attempt to modify stored data results in a hash mismatch, immediately revealing tampering.

Smart contracts enforce trust-aware policies by verifying the user's trust score before approving actions. Transactions are executed only if predefined trust conditions are satisfied; otherwise, they are blocked or flagged for review. This automated enforcement reduces reliance on manual supervision and prevents misuse of authority.

By ensuring that every modification is permanently logged and auditable, the blockchain layer prevents unauthorized or hidden changes—even by high-level administrators. Consequently, the system strengthens transparency, accountability, and protection against insider threats in digital governance environments.

VII. SECURITY ANALYSIS AND DISCUSSION

Security analysis is performed to evaluate how well the proposed Human-Trust-Aware Blockchain System protects digital governance platforms from various security threats. The focus of this analysis is on preventing silent data manipulation, detecting insider attacks, and ensuring transparency and accountability.

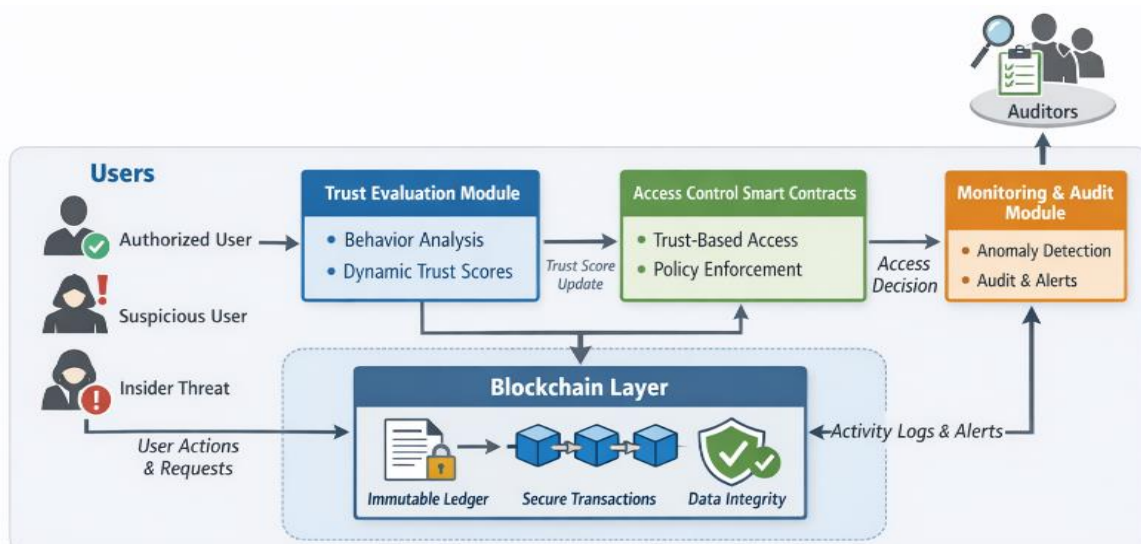


Figure 2: Security Operation Flow in the Proposed Governance System

Digital governance systems are exposed to multiple security risks, including misuse of administrative privileges, unauthorized access to sensitive public records, silent data manipulation, and lack of accountability in centralized database environments. The proposed system addresses these threats through a multi-layered security framework that integrates human trust modeling, blockchain immutability, smart contract enforcement, and continuous monitoring.

Human trust modeling continuously evaluates user behavior and dynamically updates trust scores based on operational patterns. This ensures that access privileges reflect current behavior rather than static roles. Blockchain technology provides immutable and tamper-proof storage of governance transactions, ensuring that no modification can occur without leaving verifiable evidence. Smart contracts enforce trust-based access control by validating trust conditions before approving sensitive actions, while the monitoring mechanism detects abnormal activities in real time.

Insider threats are mitigated through dynamic trust adjustment. When suspicious behavior is detected, the system automatically restricts access or generates alerts. Because all actions are permanently recorded on the blockchain, misuse becomes traceable and accountable.

Silent data manipulation is prevented by ensuring that every data operation is logged on the blockchain and cryptographically secured. Any attempt to alter stored records results in immediate detection. Additionally, trust validation occurs prior to executing sensitive operations, and audit authorities can verify transaction histories at any time, thereby maintaining transparency and accountability.

VIII. EXPERIMENTAL EVALUATION AND CASE STUDY

To test the effectiveness of the proposed Human-Trust-Aware Blockchain System, experiments were carried out in a simulated digital governance environment. The main goal was to check whether the system can prevent silent data manipulation, detect insider misuse, and maintain strong data integrity.

The testing environment included multiple authorized users with different roles, a permissioned blockchain network, a trust evaluation module, smart contracts for access control, and a monitoring system. Both normal activities and suspicious behaviors were tested to observe how the system responds under different conditions.

Different scenarios were created, such as normal role-based operations, repeated access to sensitive data, attempts to modify critical records, and unauthorized actions by insiders. The system was evaluated based on how accurately it updated trust scores, how quickly it detected threats, how well it protected data, and how effectively it maintained audit records.

The results showed that the system successfully detected suspicious behavior and reduced trust scores accordingly. Unauthorized actions were blocked before execution, and all approved operations were permanently recorded on the blockchain. No hidden or undetected data changes were observed during testing. The audit logs provided a complete and transparent record of all activities.

In a case study, an authorized official attempted to modify sensitive data beyond their allowed role. The system detected abnormal behavior, lowered the trust score, and denied the request through smart contract enforcement. Alerts were generated and securely stored on the blockchain, ensuring accountability.

Compared to traditional centralized systems, the proposed system offers stronger security by combining dynamic trust evaluation with blockchain-based record keeping. This approach improves transparency, accountability, and protection against insider threats in digital governance platforms.

IX. CONCLUSION

Digital governance platforms require strong security to protect sensitive public data from insider misuse and silent data manipulation. This paper proposed a Human-Trust-Aware Blockchain System that combines dynamic trust evaluation with blockchain-based immutable storage to enhance transparency and accountability. By continuously updating user trust scores and enforcing trust-aware access through smart contracts, the system prevents unauthorized modifications and detects suspicious behavior in real time. Experimental results and case study analysis demonstrate that the proposed approach effectively strengthens data integrity and improves overall security in digital governance environments.

X. FUTURE WORK

Future research can enhance the proposed framework by integrating advanced machine learning techniques to improve trust modeling accuracy and adaptability. Further work may focus on large-scale deployment and scalability analysis to evaluate performance in real-world governance environments. Optimizing blockchain transaction efficiency and reducing computational overhead is another important direction. The framework can also be extended to other critical domains such as healthcare and smart cities. Additionally, incorporating stronger privacy-preserving mechanisms can further improve data protection while maintaining transparency and accountability.

XI. REFERENCES

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